FakeLight

Setup:

- Import the **FakeLight** package into your Unity project
- Enable the **Depth Texture** only if using the **Geometry Fade** property
- Add the **FakeLight** component to any GameObject with or without a light component

FAQs:

- **My light isn't rendering?** - If your light currently uses the **Geometry Fade** property, either disable it or enable the depth texture if you'd like to use it, otherwise the light will become invisible
- **Does this support single-pass XR rendering?** - Yes, single-pass, single-pass instanced, multi-pass, and multiview all work for this shader
- **Does this support URP?** - Yes, all Render Pipelines are supported in this shader

Unity Asset: FakeLight
Properties Video: FakeLight Properties Video
Old Documentation: FakeLight Documentation

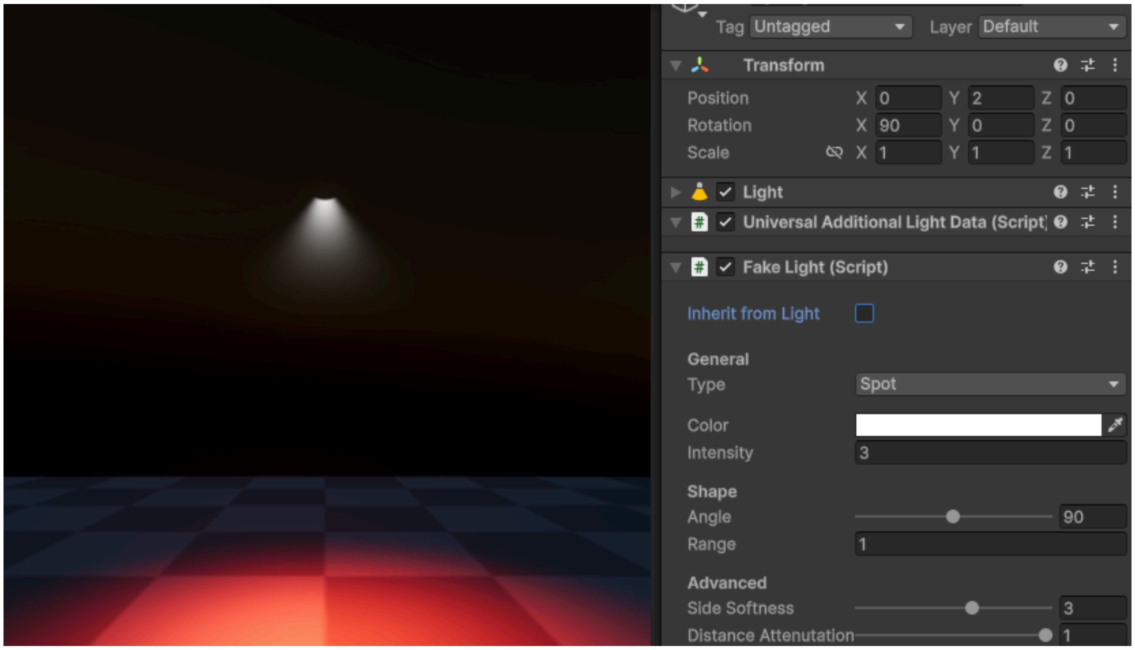
WebGL Demo
PC Demo
PCVR Demo
Meta Quest Demo

Properties:

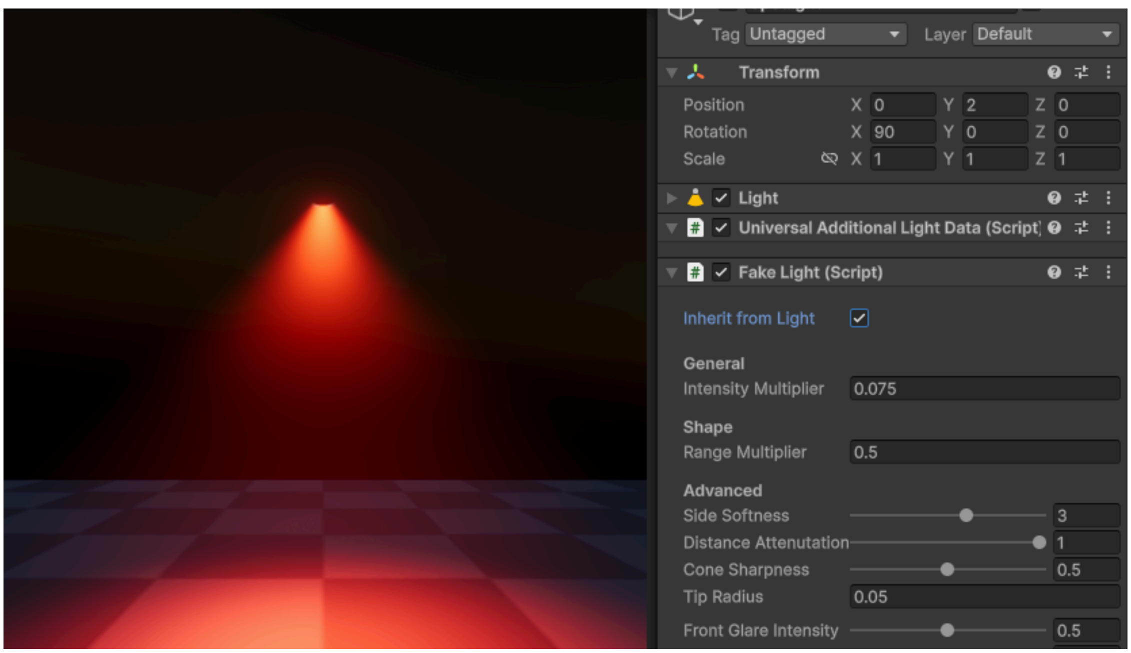
Inherit from Light - Inherit properties from a light source (color, intensity, range, spot angle). **Intensity** and **Range** turn into a multiplier property from the light source.

Intensity --> Intensity Multiplier (In **HDRP**, you may need to tune down the **Intensity Multiplier** by a significant amount due to the different light units from **Built-in** and **URP**)
Range --> Range Multiplier

Inherit from Light: **Disabled**



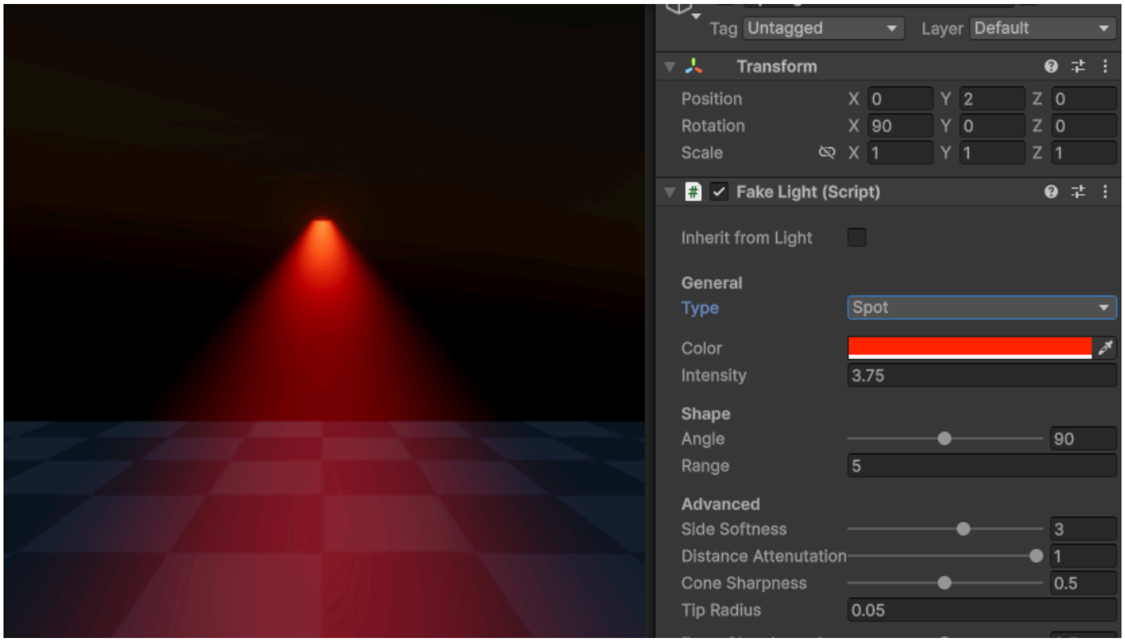
Inherit from Light: **Enabled**



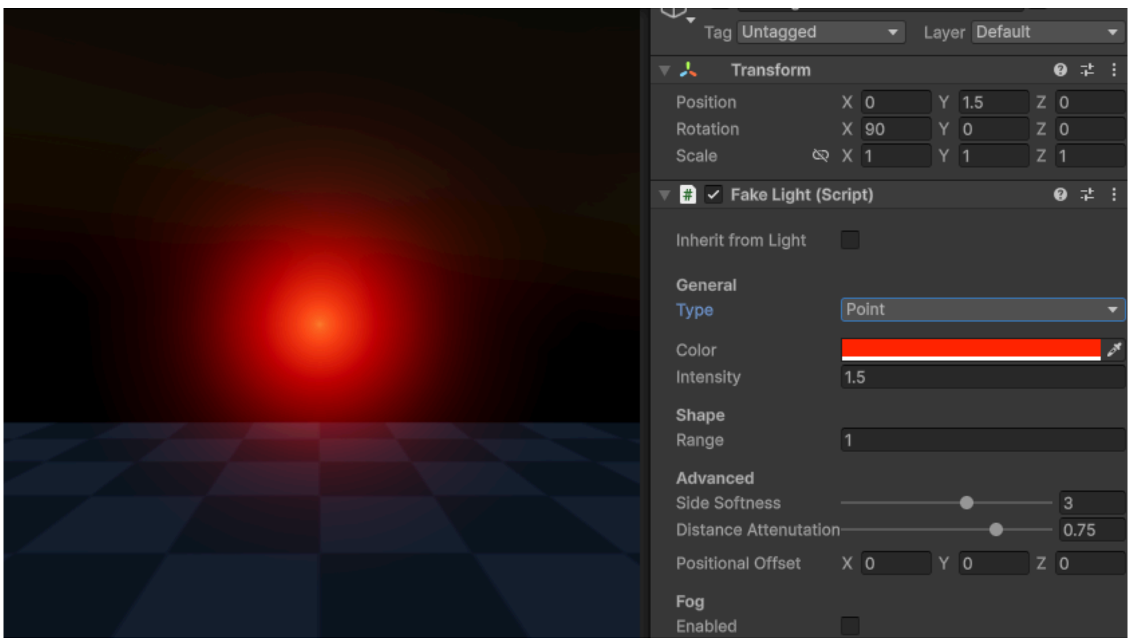
<u>General</u>

Type - Type of light (**Spot** or **Point**)

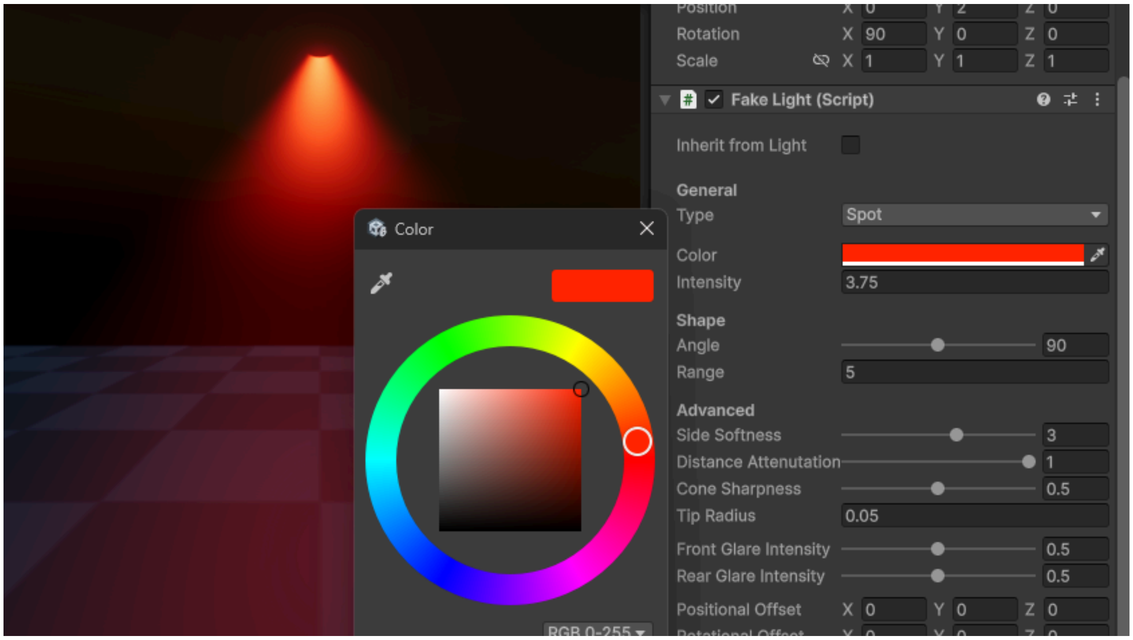
Type: **Spot**



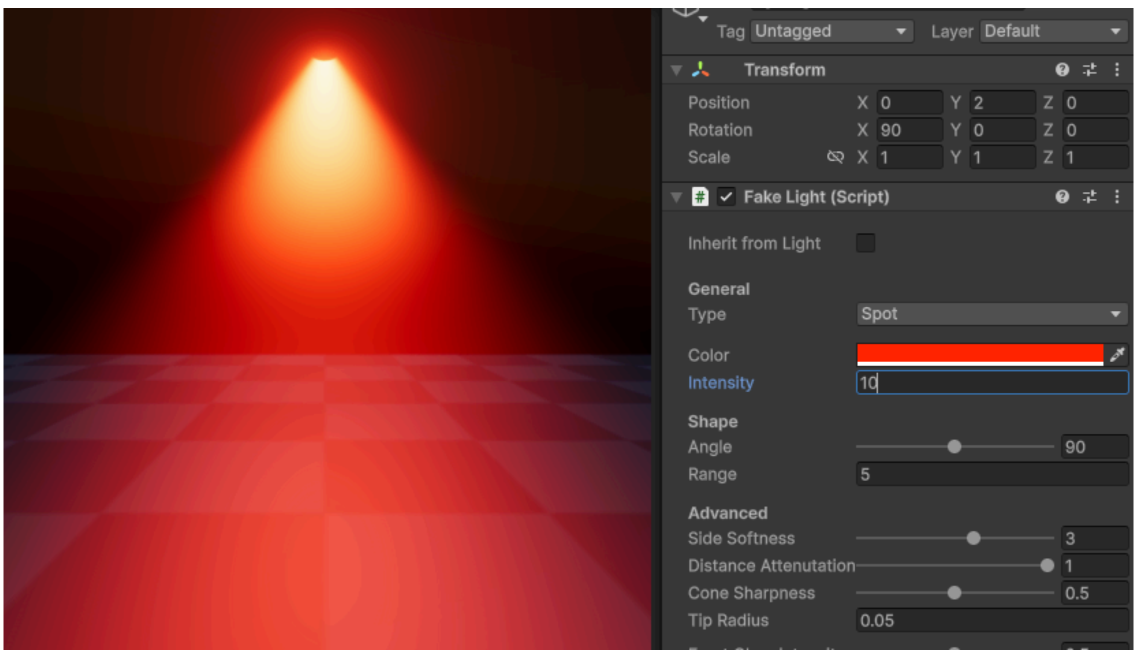
Type: **Point**



Color - Color of the light

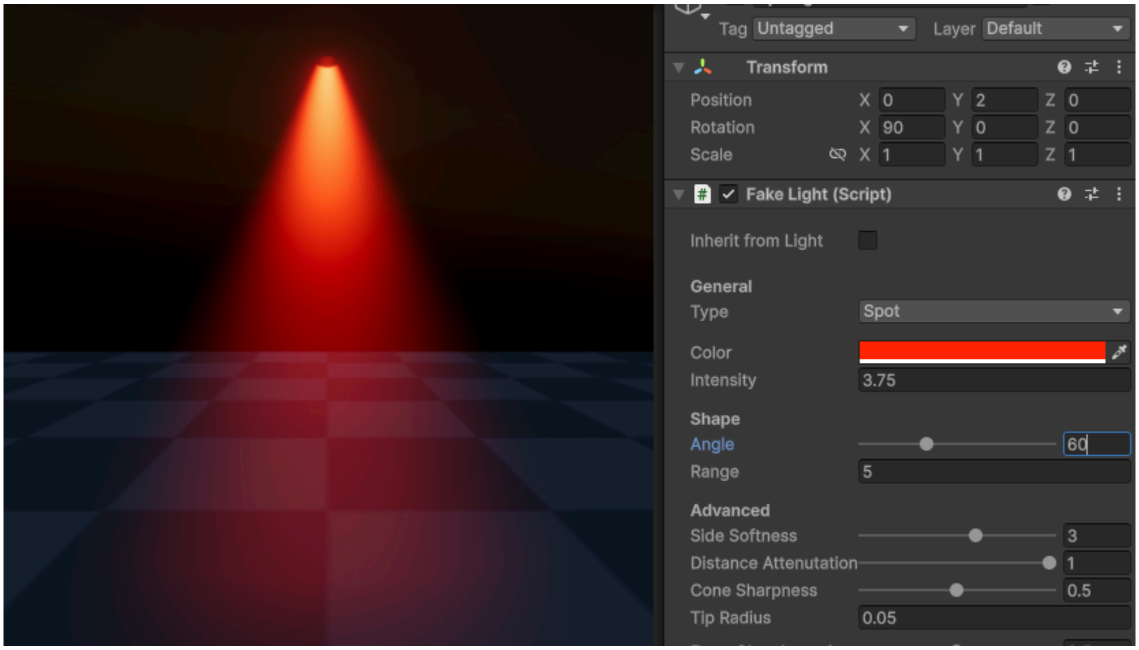


Intensity - Brightness of the light

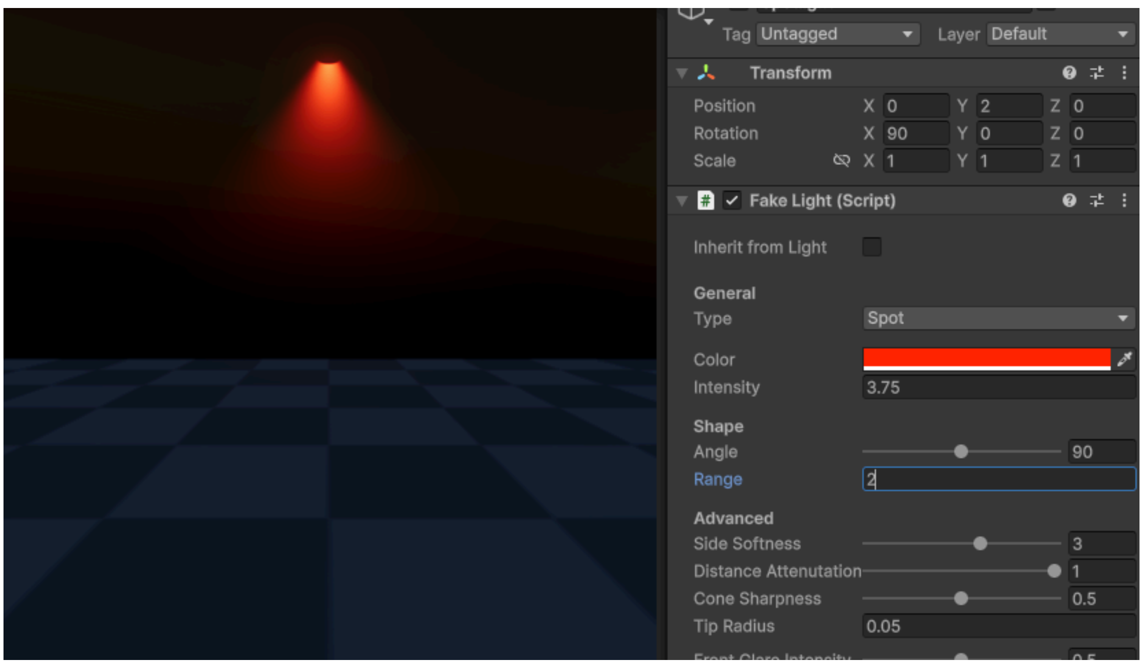


<u>Shape</u>

Angle - Angle of the spotlight in degrees

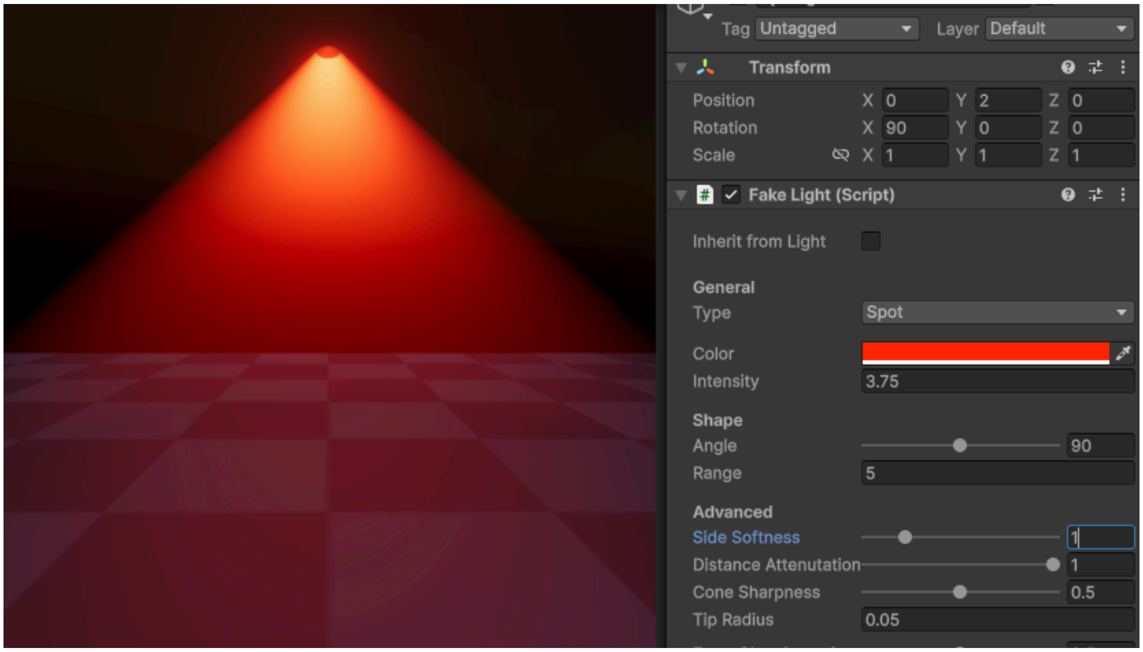


Range - Range of the spotlight in meters



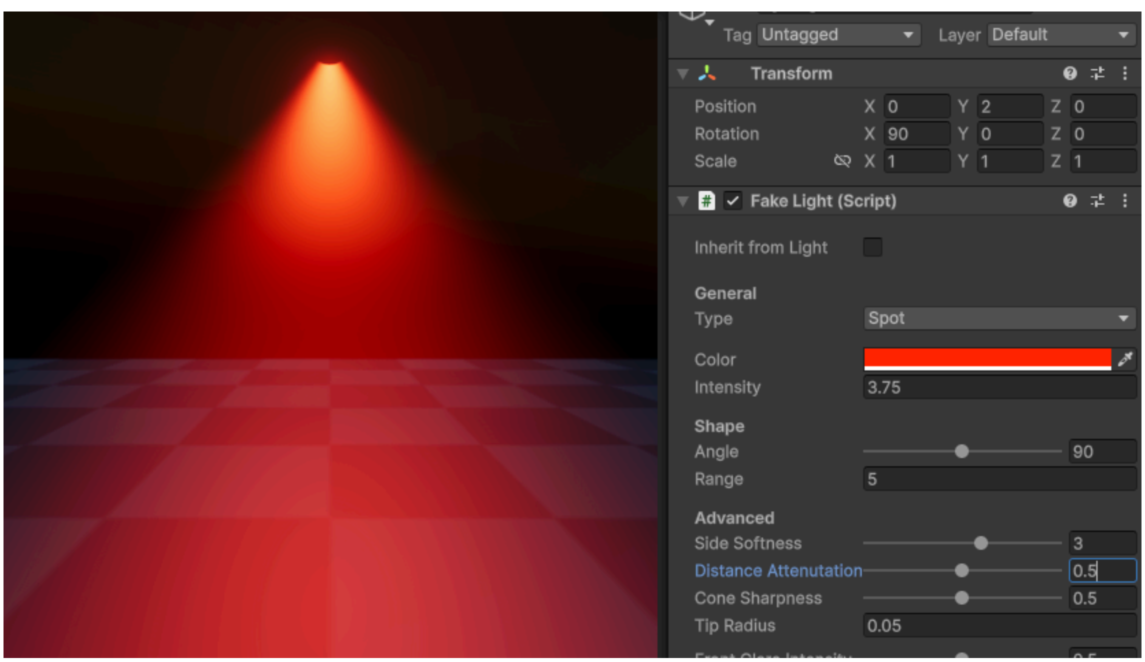
<u>Advanced</u>

Side Softness - Softness of the sides/edges

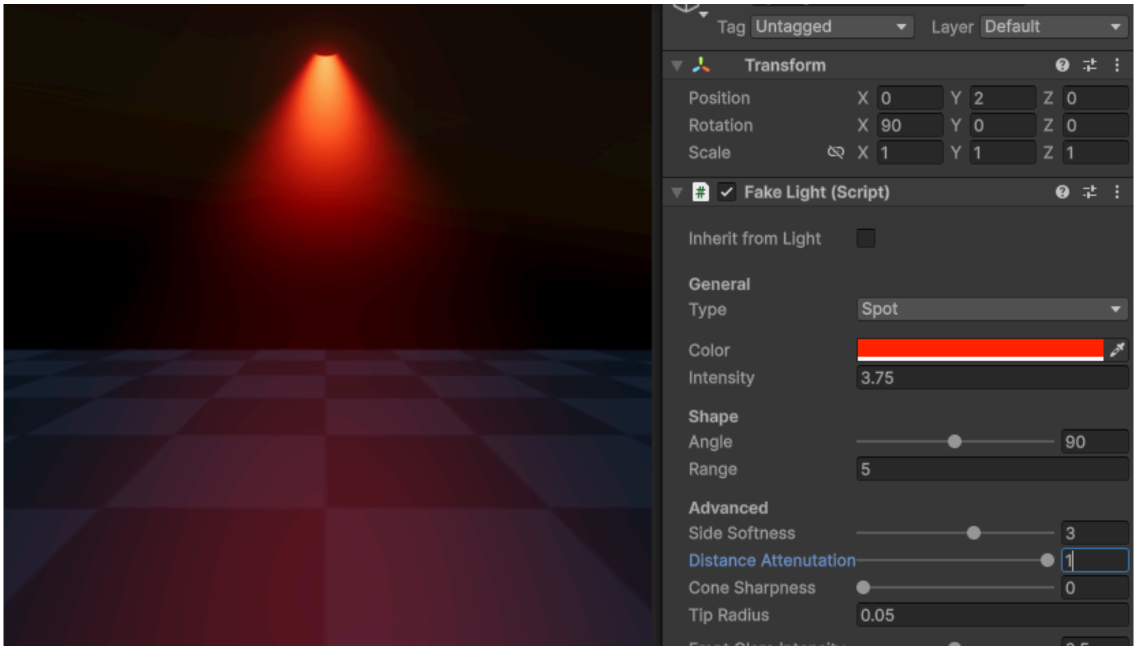


Distance Attenuation - Fall-off of light from distance

Distance Attenuation: **0.5**

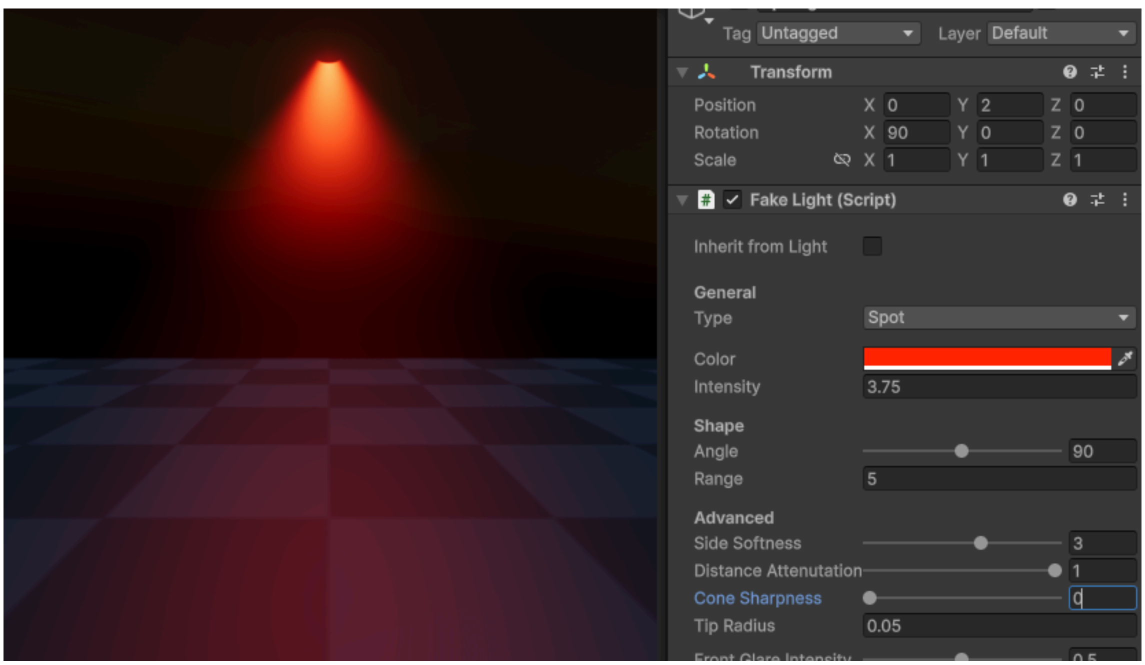


Distance Attenuation: **0**

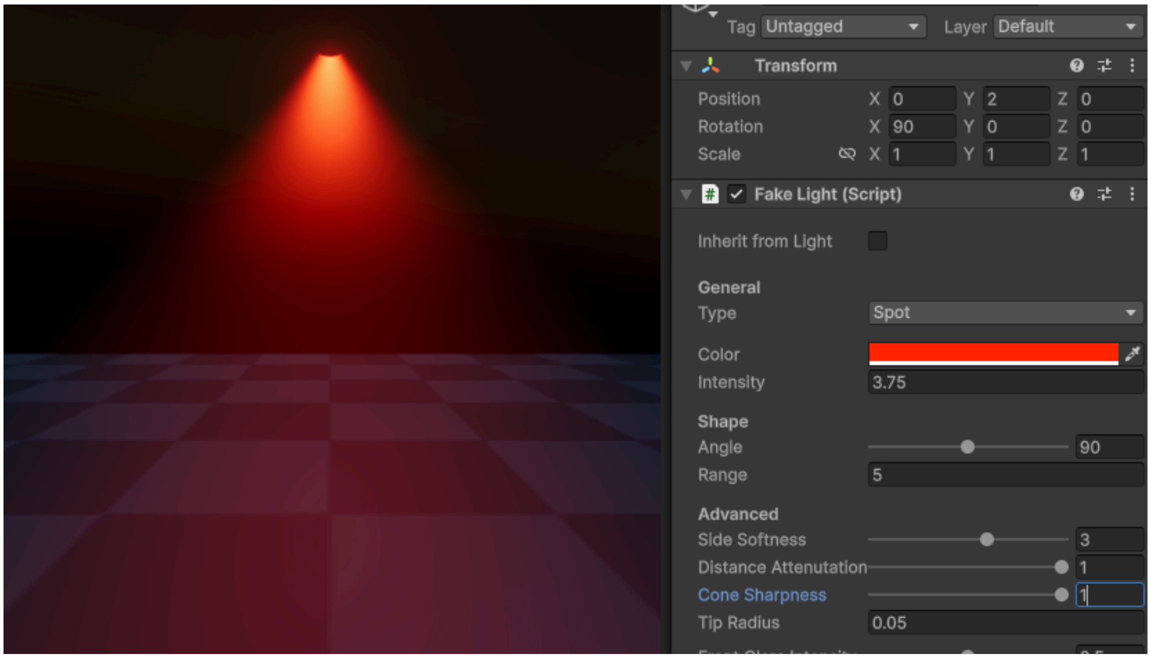


Cone Sharpness - How much to keep a cone shape

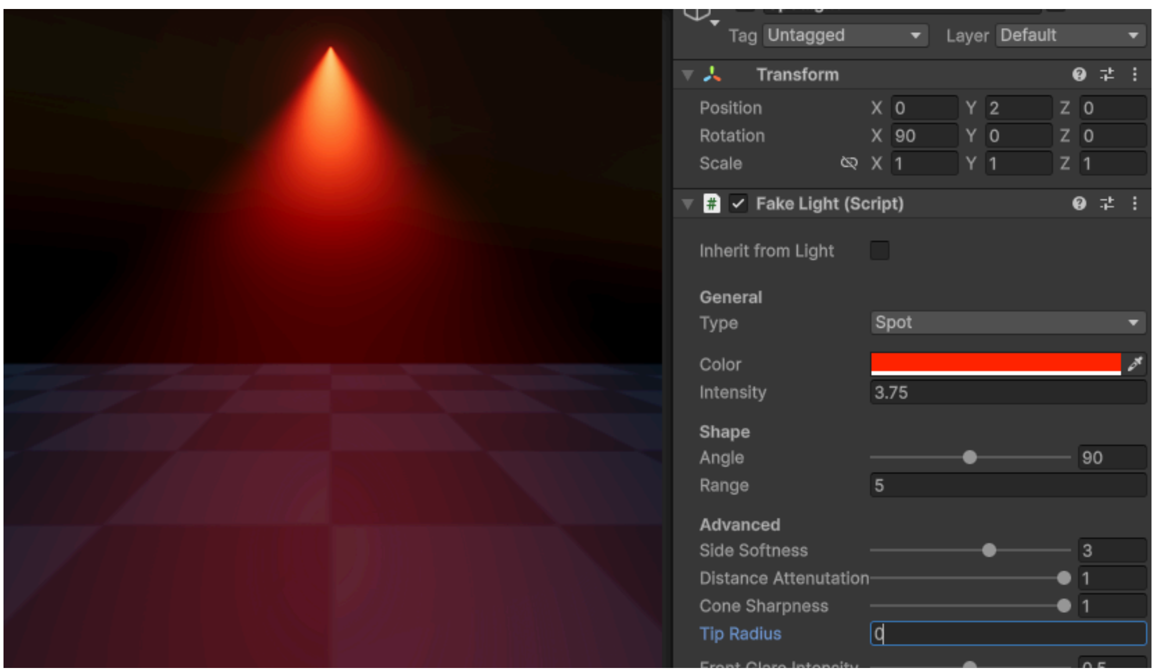
Cone Sharpness : **0**



Cone Sharpness: **1**

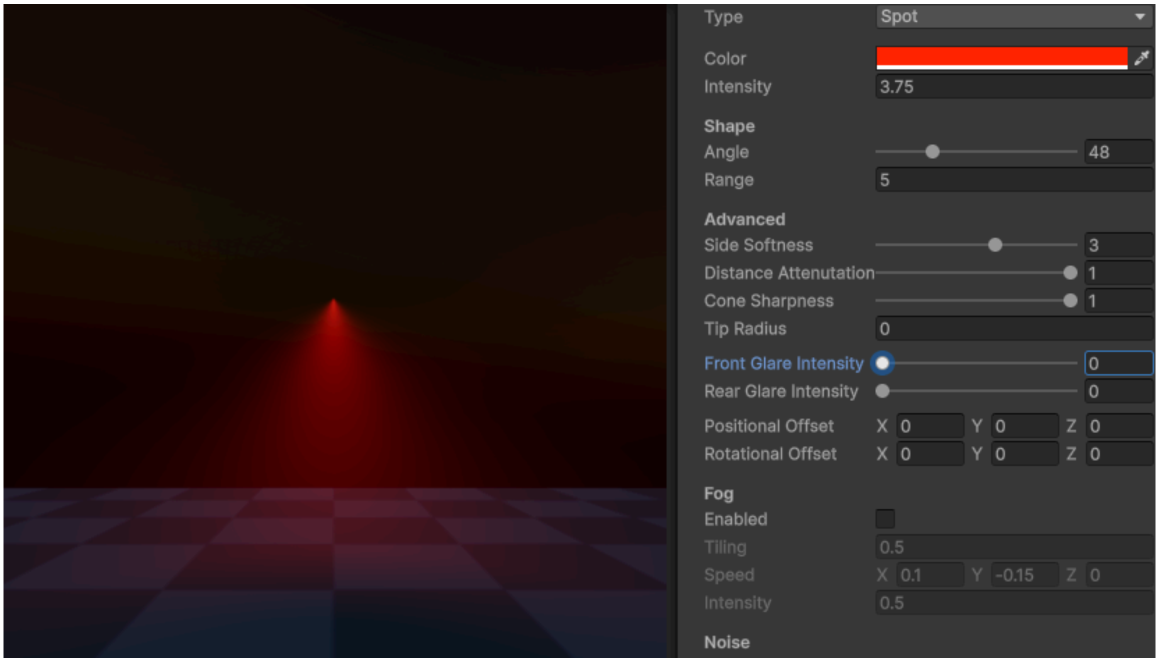


Tip Radius - Radius of cone tip.

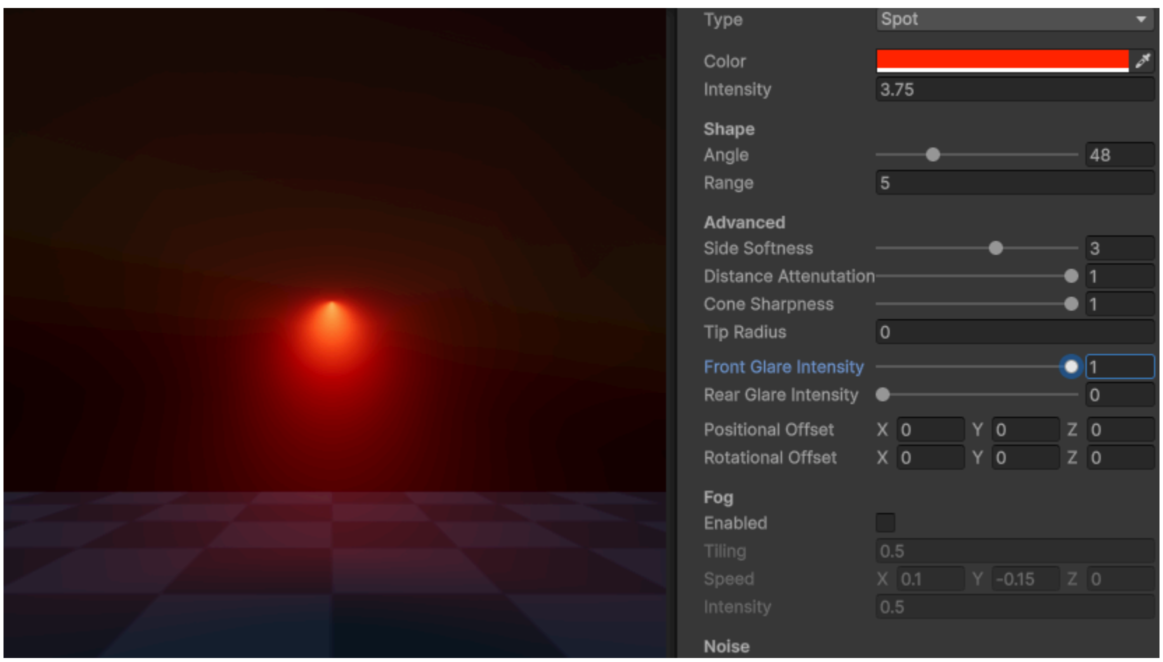


Front Glare Intensity - Boost brightness of light when looking from the front

Front Glare Intensity: **0**

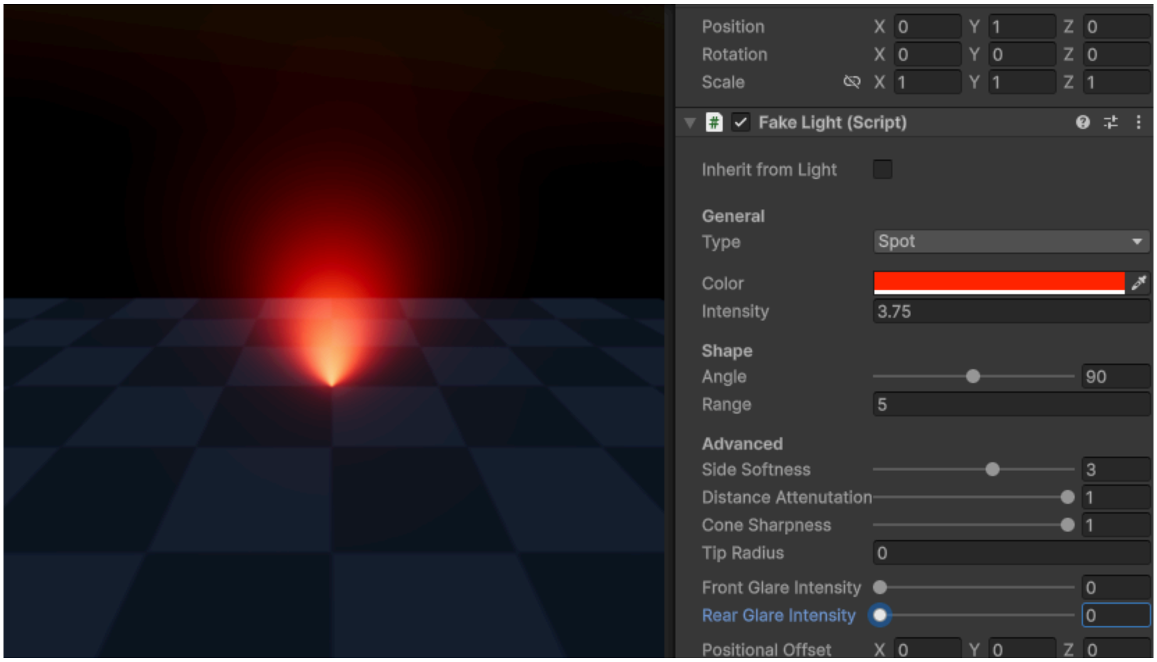


Front Glare Intensity: **1**

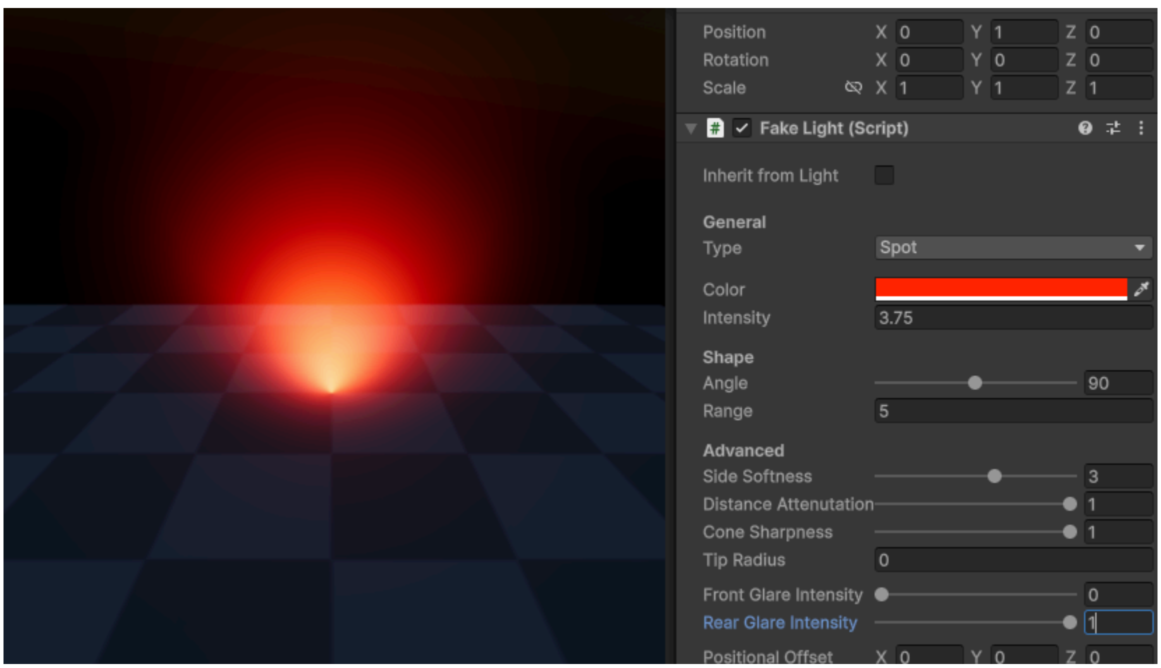


Rear Glare Intensity - Boost brightness of light when looking from behind

Rear Glare Intensity: **0**

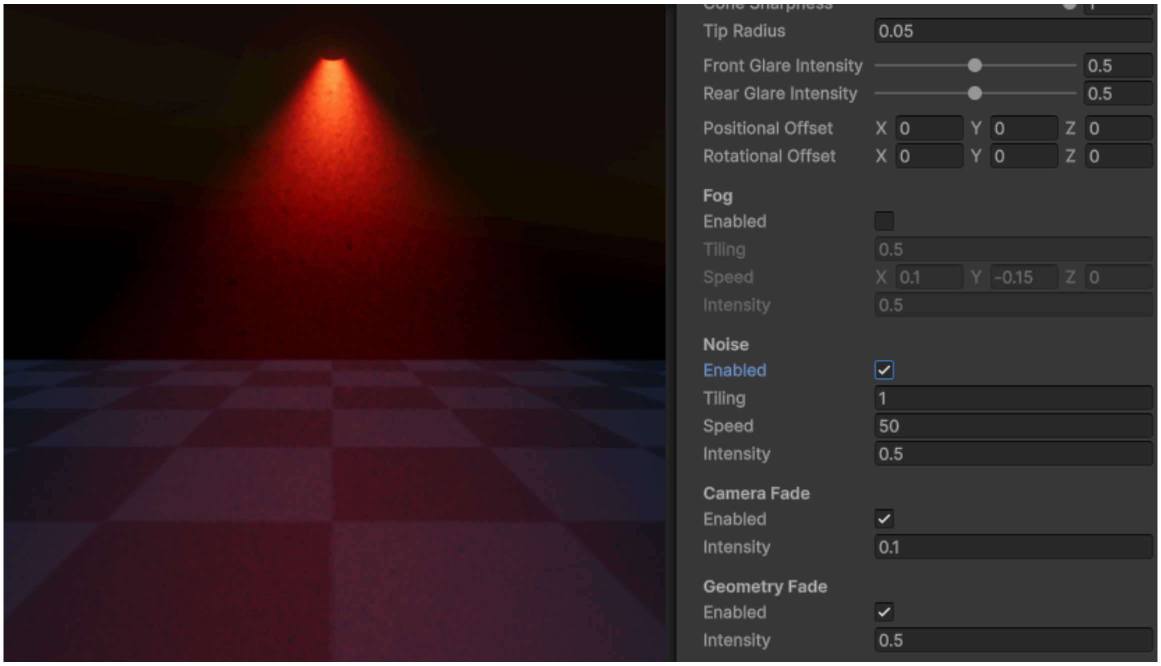


Rear Glare Intensity: **1**



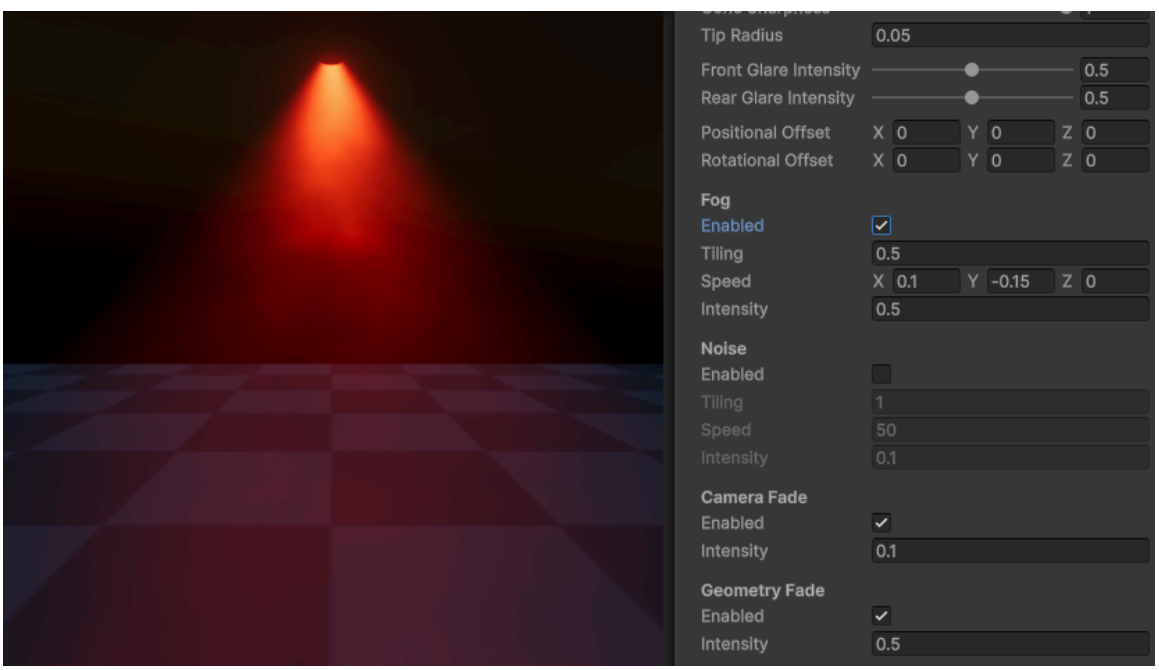
<u>Noise</u>

Enable - Enable screen space noise
Tiling - Tiling of noise
Speed - Speed of noise
Intensity - Intensity of noise



Fog

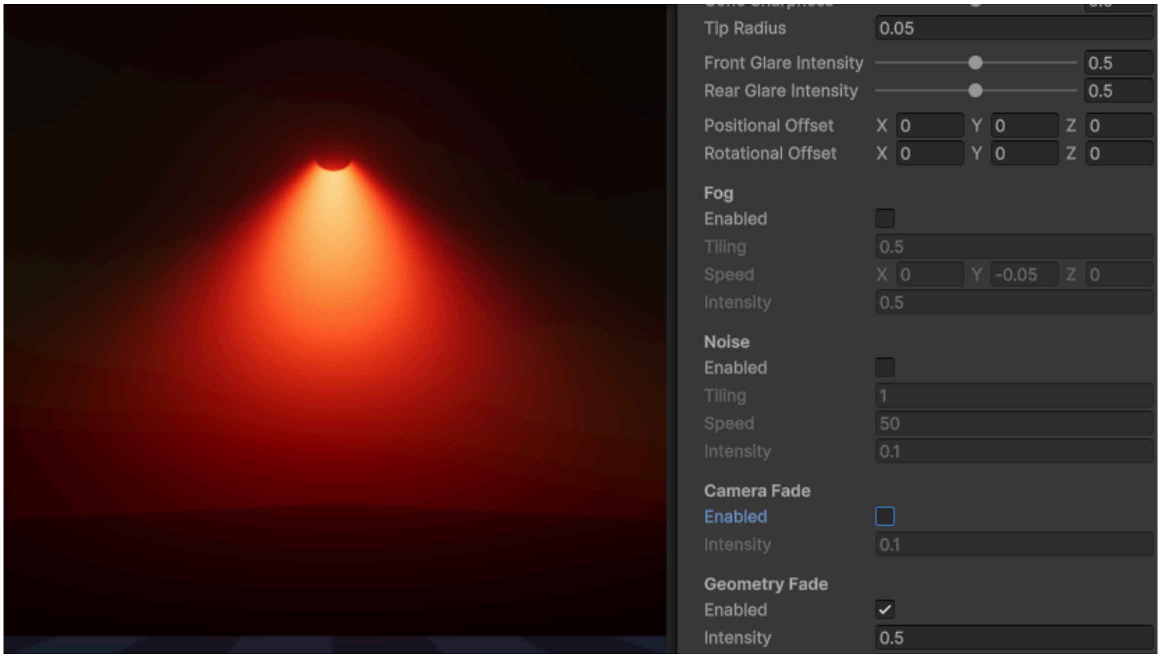
Enable - Enable world space fog
Tiling - Size of fog
Speed - Speed of fog in 3D
Intensity - Intensity of fog



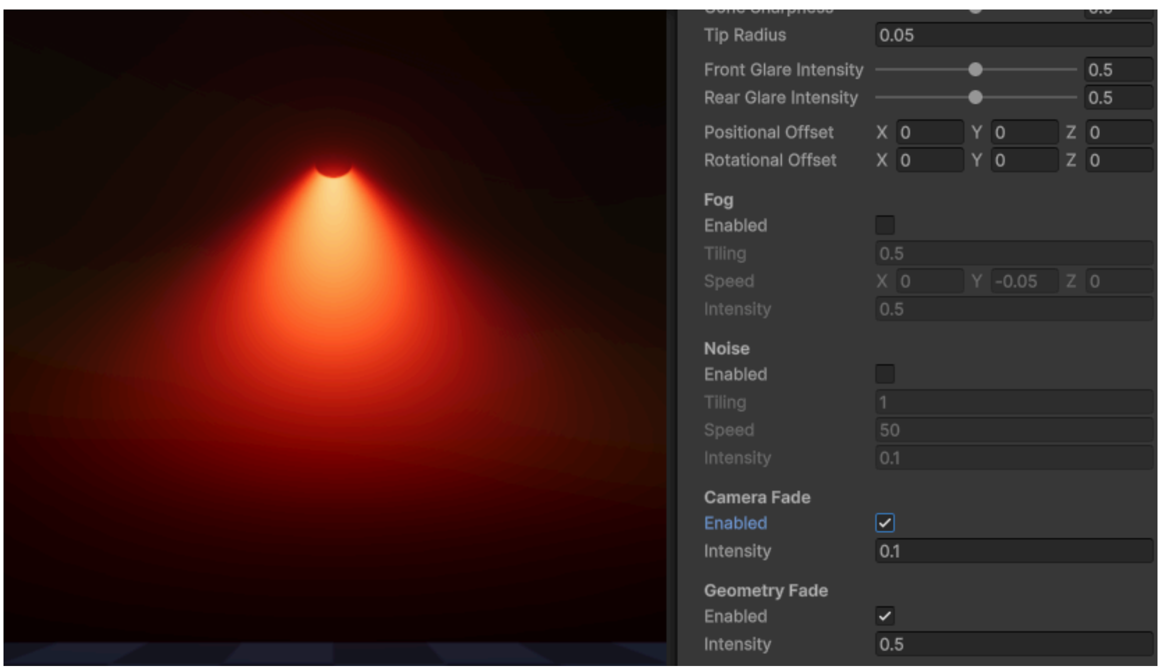
Camera Fade

Enable - Light fades as it intersects the camera, prevents camera clipping
Intensity - Intensity of camera fade

Camera Fade: **Disabled**



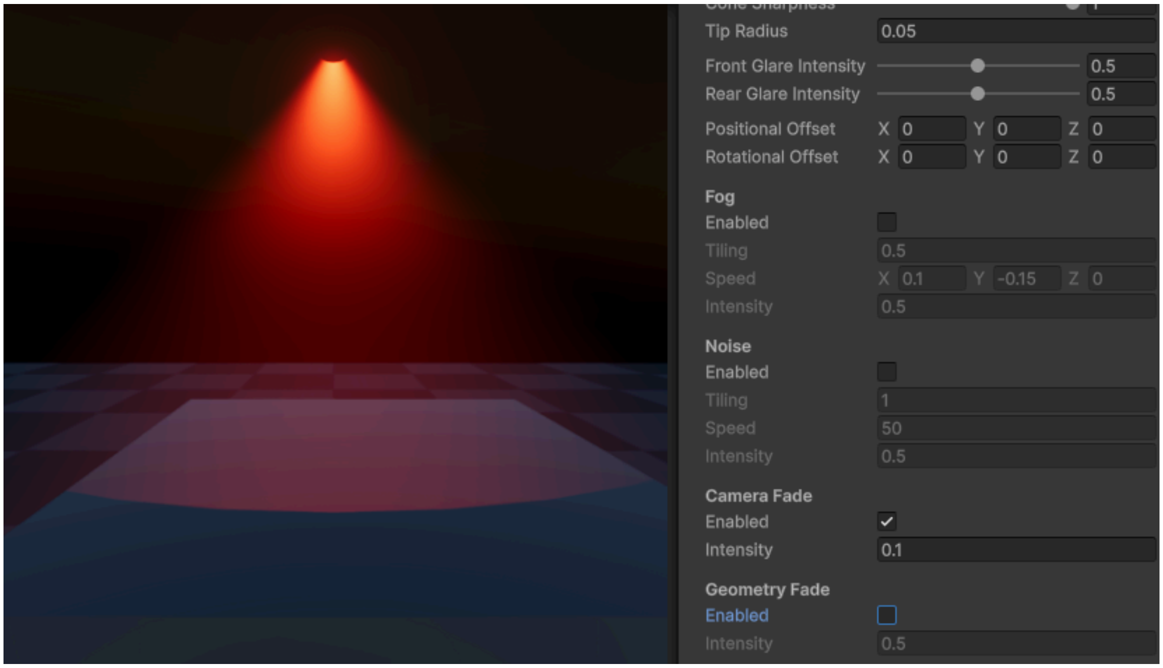
Camera Fade: **0.1**



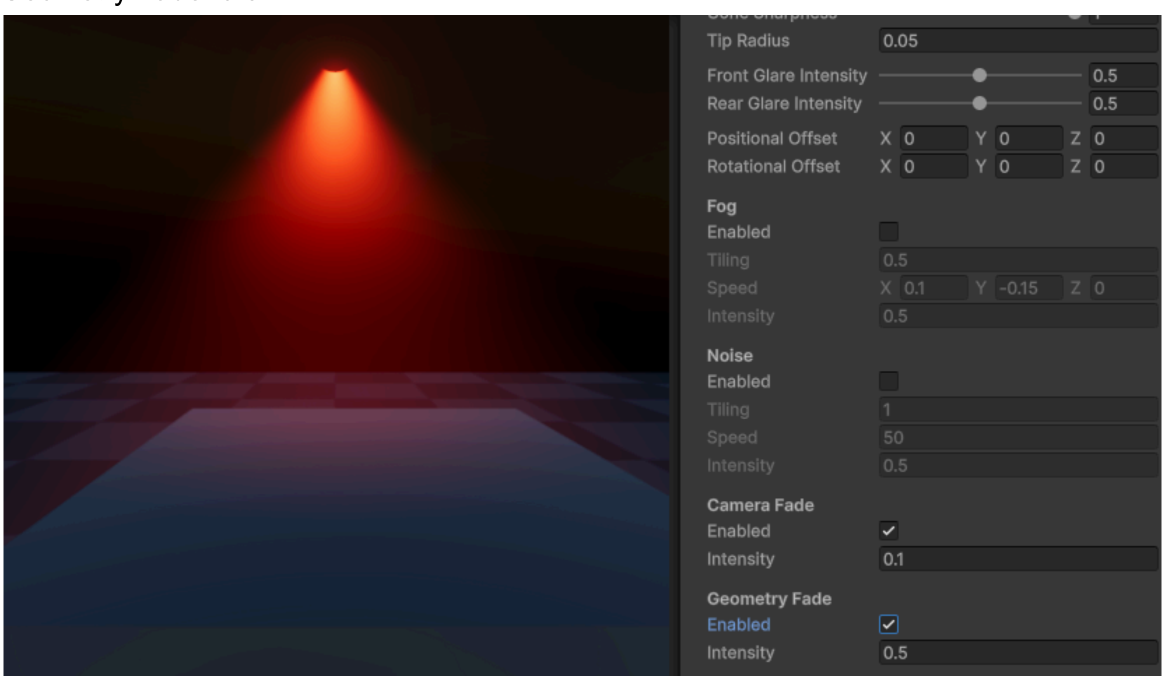
Geometry Fade

Enable - Enable geometry fade. Light fades as it intersects opaque geometry, prevents clipping (Requires **Depth Texture**)
Intensity - Intensity of geometry fade

Geometry Fade: **Disabled**



Geometry Fade: **0.5**

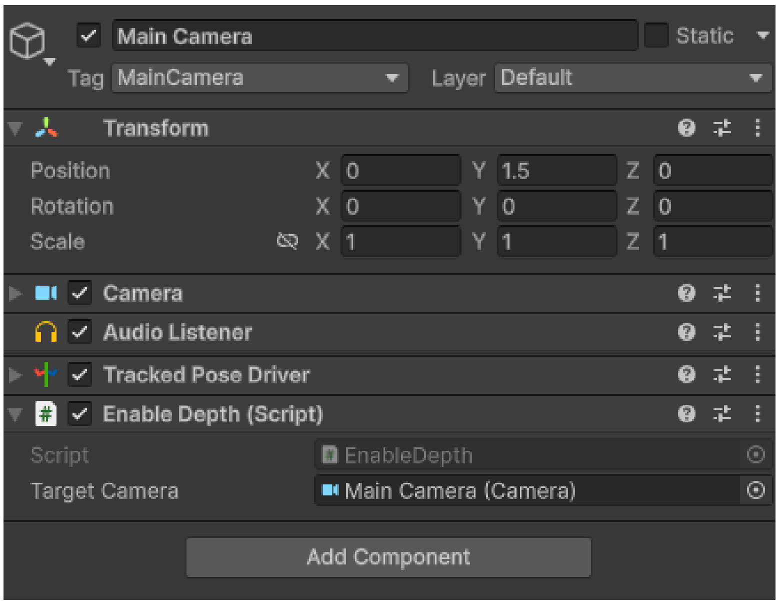


How to enable the Depth Texture in each Render Pipeline:

- Only enable **Depth Texture** if using the **Geometry Fade** property!

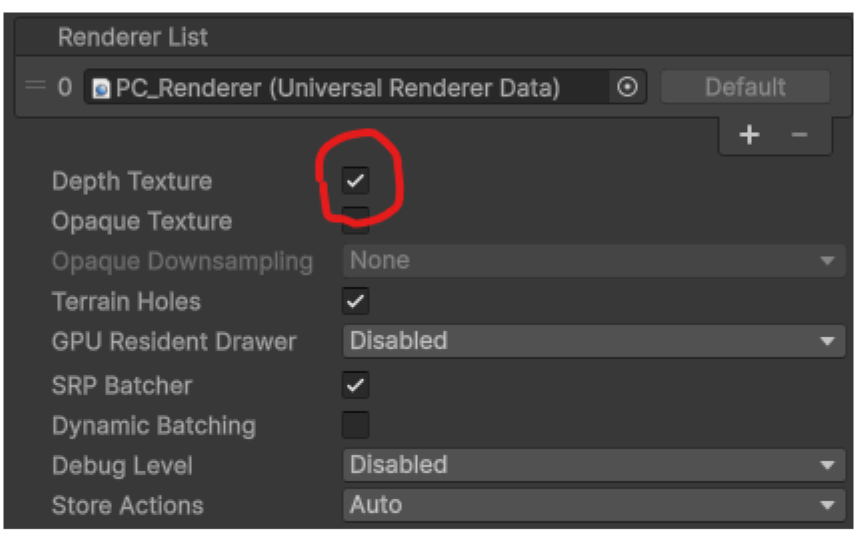
Built-in Render Pipeline

- Add the **Enable Depth** script to your current camera to enable the **Depth Texture**
- Assign the camera to the **Target Camera** property



Universal Render Pipeline

- Enable the **Depth Texture** in your current URP asset



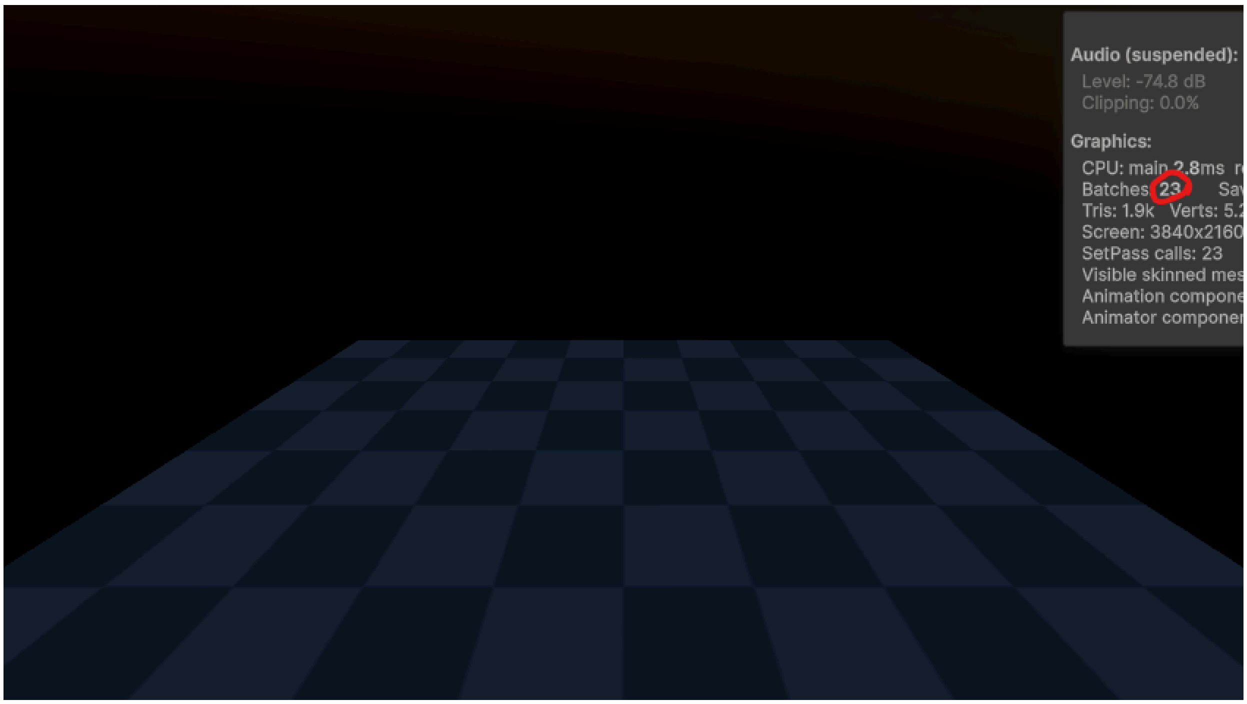
High Definition Render Pipeline

- **Depth Texture** is always enabled, no set-up required

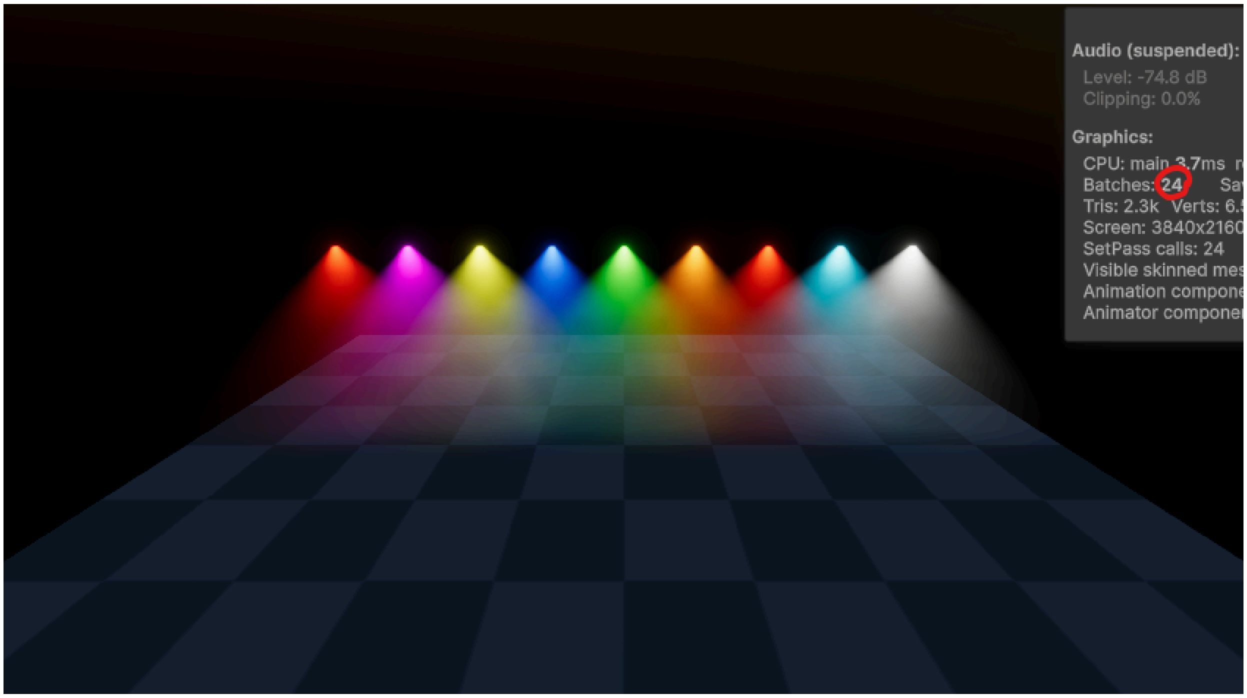
GPU Instancing

Render up to **512** lights per drawcall!

Drawcalls: **23**



Drawcalls: **24**


Audio (suspended):
Level: -74.8 dB
Clipping: 0.0%
Graphics:
CPU: main 3.7ms
Batches: 24
Tris: 2.3k Verts: 6.
Screen: 3840x2160
SetPass calls: 24
Visible skinned mes
Animation compone
Animator componer